

How to calculate quartiles

To find the values of the quartiles in a list of data, first arrange the list of numbers in ascending order from lowest to highest.

37, 38, 45, 47, 48, 51, 54, 54, 58, 60, 62, 63, 63, 65, 69, 71, 74, 75, 78, 78, 80, 84, 86, 89, 92, 94, 96

Using the formulas, calculate where to find the quartiles and the median in this list. The answers give the position of each value in the list.

n is the total number of values in the list

$$\frac{(n+1)}{4} = \frac{(27+1)}{4} = 7$$

formula to find lower quartile

△ **Lower quartile**
This calculation gives the answer 7, so the lower quartile is the 7th value in the list.

position of lower quartile (7th value)

$$\frac{(n+1)}{2} = \frac{(27+1)}{2} = 14$$

formula to find median

△ **Median**
The answer to this calculation is 14, so the median is the 14th value in the list.

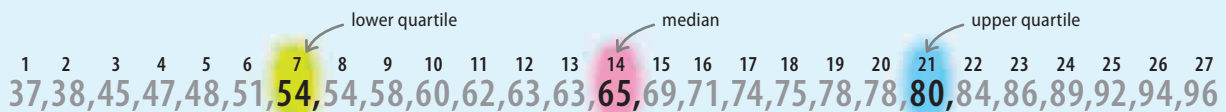
position of median (14th value)

$$\frac{3(n+1)}{4} = \frac{3(27+1)}{4} = 21$$

formula to find upper quartile

△ **Upper quartile**
The answer to this calculation is 21, so the upper quartile is the 21st value in the list.

To find the values of the quartiles and the median, count along the list to the positions that have just been calculated.



LOOKING CLOSER

Box-and-whisker diagram

Box-and-whisker diagrams are a way of showing the spread and distribution of a range of data in an graphic way. The range is plotted on a number line, with the interquartile range between the upper and lower quartiles shown as a box.

▽ Using the diagram

This box-and-whisker diagram shows a range with a lower limit of 1 and an upper limit of 9. The median is 4, the lower quartile 3, and the upper quartile 6.

